

What does the edge cost? Capital-adjusted market making under a public portfolio-margin engine

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ABSTRACT

A market maker quotes against a budget of capital, not contracts, yet its edge is reported per contract. On Derive, a crypto options exchange, the margin rules are public smart contracts, and its off-chain engine quotes margin for any book. Capital is what its three margin managers require for positions, not balances; it is measured on one contract for each fill of a companion paper and on the books of dominant maker subaccounts. Four hypotheses were registered before the analysis's capital figures existed. Contrary to H1, capital does not reorder the money-ness-by-tenor map of the edge in the registered sense (rank correlation 0.890); post hoc, much of the correlation comes from the sign of the edge. In the books of the four subaccounts under the second-generation portfolio manager (PM2), trading only ETH and HYPE, the median fill binds 3.5 per cent of its stand-alone capital (H2 not rejected). On the opening books of nine dominant subaccounts, standard margin needs 4.73 times the PM2 capital (H3 not rejected), mostly for books too large for a standard-margin subaccount. Contrary to H4, cheaper capital did not detectably narrow the half spread; the test has little power. The sample ends on 30 September 2026.

1. Introduction

A market maker on an options venue does not run out of contracts. It runs out of capital. Funding constraints act on the deposit that backs the margin of its positions (Brunnermeier and Pedersen, 2009), and margin-based pricing turns that deposit into a required return (Gârleanu and Pedersen, 2011). The literature on market making nonetheless counts in contracts or their value. Inventory models penalise the position (Ho and Stoll, 1981; Avellaneda and Stoikov, 2008) or bound it in units (Guéant et al., 2013). Option market making prices the risk of the book (Stoikov and Sağlam, 2009; Gârleanu et al., 2009), the spread compensates for risks the maker cannot hedge (Jameson and Wilhelm, 1992), and inventory risk moves option prices (Murav'yev, 2016). Even the illiquidity premium in options is measured against the option price (Christoffersen et al., 2018). Capital enters option prices through the maker's wealth (Fournier and Jacobs, 2020) or aggregate intermediary constraints (Chen et al., 2019). What a unit of edge costs in capital remains open: outsiders can rarely combine an exact margin rule with the books makers actually hold.

Derive, a crypto options exchange, makes both observable. It margins every subaccount under one of three managers: a strategy-based standard manager (SM), a legacy portfolio manager (legacy PM) and a second-generation portfolio manager (PM2). Each is a deterministic function of the positions, the price feeds and risk parameters stored on chain, and its smart contracts can be queried at any past block. A public endpoint evaluates any hypothetical book with the venue's off-chain trading engine. Makers' holdings are on chain too, so their actual books can be priced. Capital is measured on positions under the venue's rules, not on

account balances, which mix what a book needs with what a trader deposits.

The question extends Albiez (2026), which maps the maker's net edge over money-ness and tenor. Here the same fills are priced in capital: how much capital does the edge bind, and does the map change when the edge is measured per unit of capital? Option strategies lose much of their appeal once margin is charged (Santa-Clara and Saretto, 2009); this paper takes the maker's side and uses the rule the venue actually applies.

The paper makes four contributions: a capital map over the money-ness and tenor grid of Albiez (2026); the marginal cost of a fill in the actual book of a dominant subaccount; the value of netting on the same books under standard margin and PM2, in the tradition that compares strategy-based and portfolio margining on identical positions (Kupiec and White, 1996); and a price of capital. Makers' inventories and revenues move liquidity (Comerton-Forde et al., 2010), and margin changes move swaption prices (Ahn, 2025); here, parameter changes form a quasi-experiment: on-chain events that change the stand-alone capital of every cell by a known amount.

Four hypotheses were registered before any capital figure of the registered analysis existed: that edge per notional and edge per unit of PM2 capital rank the cells differently (H1); that a fill in the book of a dominant subaccount under PM2 binds less than half its stand-alone capital per contract (H2); that PM2 saves more than half the capital standard margin requires for the same books (H3); and that the half spread falls when a parameter change makes capital cheaper (H4). Addenda 1 to 7 of the pre-registration are described before the appendices.

H1 and H4 are rejected; H2 and H3 are not. The capital denominator does not reorder the map in the registered sense: the rank correlation is 0.890, with an interval from

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BTC · 17 Sep 2026, 08:00 UTC · 15 live expiries · PM2 spot grid $\pm 14\%$ since 20 Aug 2026 · on-chain engine

a PM2: ATM 30 d short = 11.8 % of forward

b SM: ATM 30 d short = 14.1 % of forward

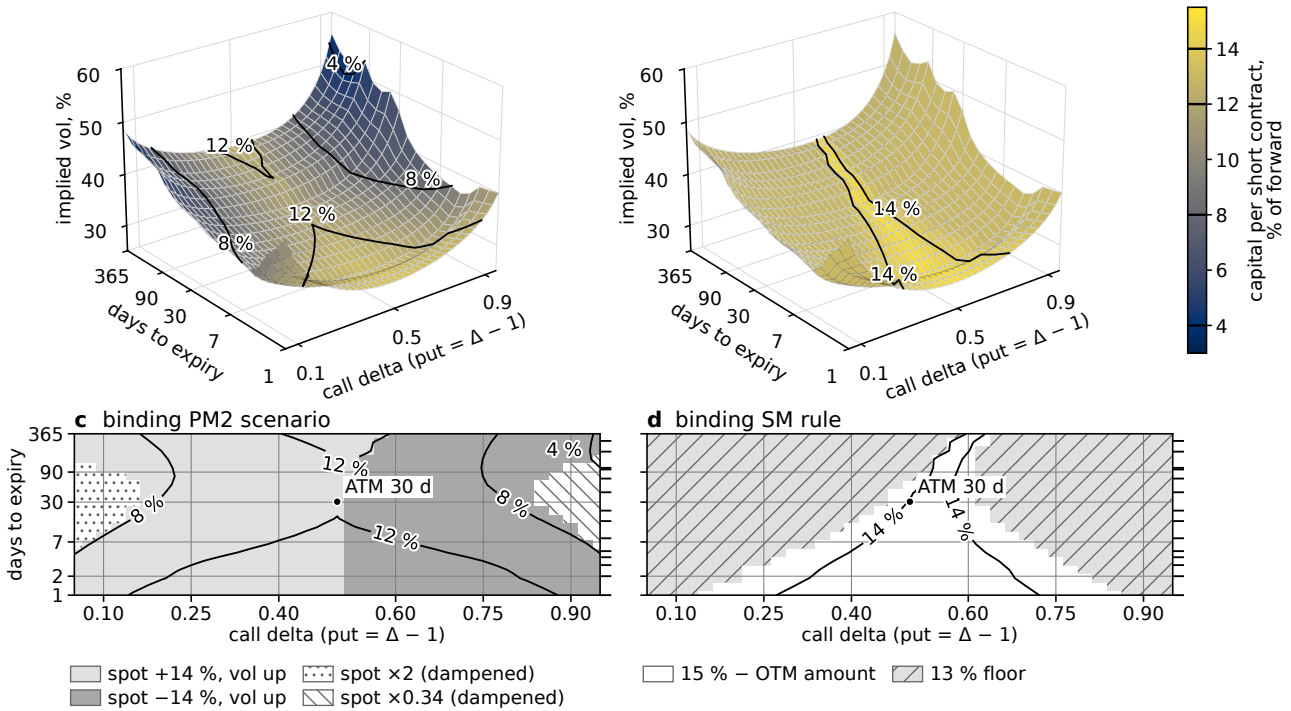


Figure 1: The surface as the engine sees it. BTC implied volatility over call delta and tenor at 08:00 UTC on 17 September 2026, built from the on-chain feeds, with the capital that one short contract binds under PM2 (panel a) and under standard margin (panel b) as colour on a common scale in per cent of the forward. Height is volatility, colour is capital, and black lines join points of equal capital. ATM marks the at-the-money node at 30 days to expiry. Panels c and d name the rule that sets the capital at each point: the worst scenario of the PM2 grid, and the branch of the standard margin formula, a share of spot less the out-of-the-money (OTM) amount or a floor. Their grid lines are the tenor and absolute delta bucket edges of Figures 3 and 4, and the ticks on their right edge mark the listed expiries. For puts the delta edges are mirrored: the lines at 0.60, 0.75 and 0.90 are the put edges of 40, 25 and 10 per cent. The surface holds out-of-the-money options only, so the buckets from 60 per cent absolute delta up have no counterpart here. Standard margin is set in per cent of spot and shown in per cent of the forward. Capital follows the smart contracts on chain; the venue's off-chain engine discounts PM2 at a flat two per cent.

0.883 to 0.906 against a threshold of 0.5 (H1). Post hoc, much of this correlation comes from the sign of the edge: among profitable cells it is 0.630 (0.585 to 0.700). In the books of the four subaccounts under PM2, which trade ETH and HYPE only, the median fill binds 3.5 per cent of its stand-alone capital (H2); maker buys there mostly release capital, sells mostly bind it (post hoc). Standard margin needs 4.73 times the PM2 capital for the opening books of nine dominant subaccounts (H3), most of these books larger than a standard-margin subaccount accepts. I find no evidence that the half spread fell when capital got cheaper (H4): the estimate has the wrong sign and a one-sided p of 0.62, but the test cannot exclude a sizeable narrowing.

2. The engine and what it returns

This section derives the capital measure from the margin rules. The standard manager is strategy based. Each option is margined in isolation, each expiry takes the smaller of its summed option margins and its maximum loss, and expiries add up. A long option earns no credit of its own,

so under SM a long book's capital is its full premium. The legacy manager and PM2 are scenario based, in the spirit of Standard Portfolio Analysis of Risk (Kupiec, 1994): the book is revalued under spot and volatility shocks, and the requirement is the worst loss, scaled by a margin factor, plus contingencies (panel c of Figures 1 and 2). Both haircut shocked expiry values with a static discount. PM2 adds skew scenarios, applies the haircut to positive and negative expiry values alike, and discounts at a live rate feed; the legacy manager's rate stays at zero. Both net within one underlying and never across, and the scope of netting decides what it saves (Duffie and Zhu, 2011; Cont and Kokholm, 2014). Probability-based margin goes back to Figlewski (1984), and the worst-loss core of scenario margin is a coherent risk measure (Artzner et al., 1999). The managers' smart contracts are public (Derive, 2026b), with parameters on chain as in the lending protocols of Qin et al. (2021); BitMEX instead margins each position at a percentage of its notional (Soska et al., 2021).

The public endpoint `get_margin` (Derive, 2026a) does not return a requirement. For positions q and USDC collateral C

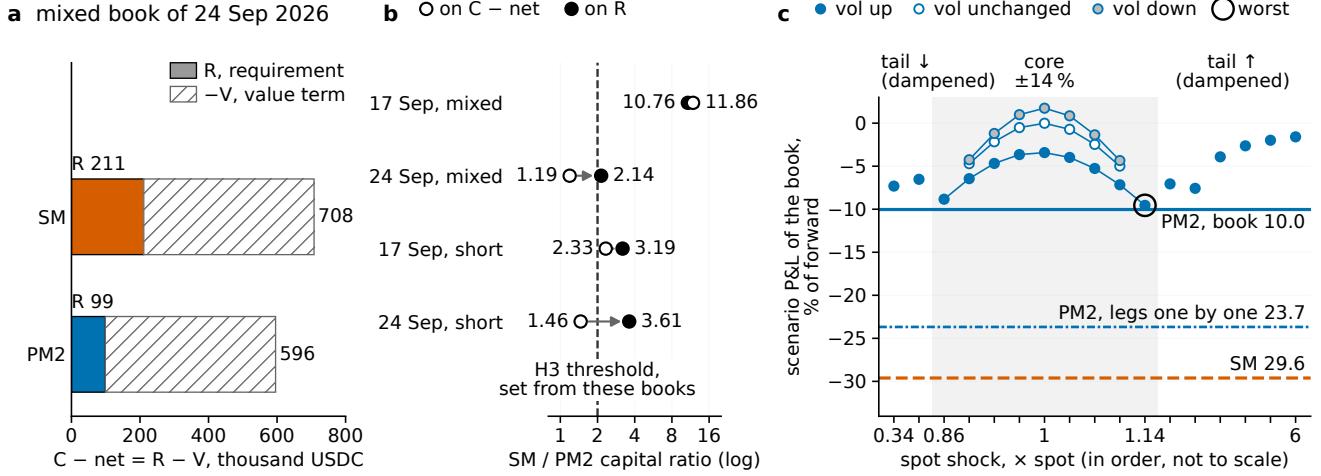


Figure 2: What `get_margin` returns, and how PM2 prices a book. Panel a splits $C - \text{net}$ into the requirement R and the value term $-V$ for the mixed probe book of 24 September 2026 (block 45 110 142); $-V$ is almost the same under both managers and dwarfs R . Panel b sets the ratio of standard to PM2 margin read as $C - \text{net}$ (open) against the ratio on R (filled) for the four probe books at their historical blocks. A short book is short in every option it holds, a mixed book long in some and short in others. The dashed line is the H3 threshold of two, which was set from these books before any maker book was measured. On the mixed book of 17 September, V is positive, so the ratio falls from 11.86 to 10.76. Panel c is the scenario profit and loss of a short BTC straddle struck at the forward on 17 September 2026, on the listed expiry nearest to 30 days (22 days), in per cent of the forward, over the spot shocks in order, not to scale. PM2 charges the worst scenario of the whole book plus contingencies (solid line), which is less than the sum of the legs margined one by one (dash dot); standard margin is dashed.

it returns, for initial and for maintenance margin X ,

$$\text{net}_X(q; C) = C + V(q) - R_X(q), \quad (1)$$

with V the value of the positions in the manager's own valuation and R_X the requirement. The venue presumably admits a book whose initial net margin is non-negative; the chain enforces a weaker check, so binding initial margin is an assumption. The fields `pre` and `post` evaluate the book as given and after simulated changes at one price state within one request, which makes a marginal cost measurable.

Treating $C - \text{net}$, which equals $R - V$, as the requirement would mix the value V into the margin. Before the registration, this gave ratios of standard to PM2 margin of 11.86 and 1.19 for the mixed probe books of 17 September and 24 September 2026; on R the ratios are 10.76 and 2.14 (Figure 2). The second book's PM2 value of $-496\,859$ USDC adds almost the same amount to numerator and denominator, pulling the ratio towards one. Netting factors are properties of books, so this paper measures them on real ones.

The quantity used throughout is the capital needed to open a book q at traded prices p and meet initial margin,

$$K_p(q) = \sum_{i \in \text{options}} p_i q_i - \text{net}_{\text{IM}}(q; 0). \quad (2)$$

Collateral enters net margin one for one, so K_p is the deposit at which the book is exactly admissible once the premium has changed hands. The measure needs no valuation of its own: it differs from the requirement R_{IM} by $\sum p_i q_i - V$, a valuation convention, not capital. For a trade Δq at prices

p , one request with a collateral change of $-p \Delta q$ gives $\Delta K = -(\text{post} - \text{pre})$. Figure 1 maps K_p of single contracts on the BTC surface.

Two engines implement the managers. The off-chain engine behind the endpoint, on which the venue trades, prices at fresh ticker marks and discounts PM2 at an undocumented flat rate; a box spread in each of the fourteen listed BTC expiries on 24 September 2026 puts that rate at 2.0000 per cent. The on-chain engine settles and liquidates, reads feeds that lag the ticker, and discounts with the rate feed, which ranged from 3.64 to 3.82 per cent across the same expiries. Only the on-chain engine can be rebuilt for past blocks, so past fills are priced at their block with offline replicas of its smart contracts. On 195 single contracts on 25 September 2026, the two engines differed in PM2 capital by a median of 0.08 per cent and at most 2.4 per cent, at the longest tenor; whole books, probed at random, showed larger gaps.

3. Data and measurement

The fills are those of Albiez (2026): all 621 508 BTC, ETH and HYPE option fills from 11 January 2024 to the cut-off of 30 September 2026. Each carries the maker's side, price, size and net edge in USDC per contract: the markout against the mark thirty minutes after the fill, less fee net of rebate and a hedging cost. Fee and rebate, booked per fill, are divided by its size (Addendum 2). On a multi-leg request for quote (RFQ) the venue almost always books the package fee on one leg, where it stays (Addendum 6). The companion paper was corrected on 25 September 2026 and uses the

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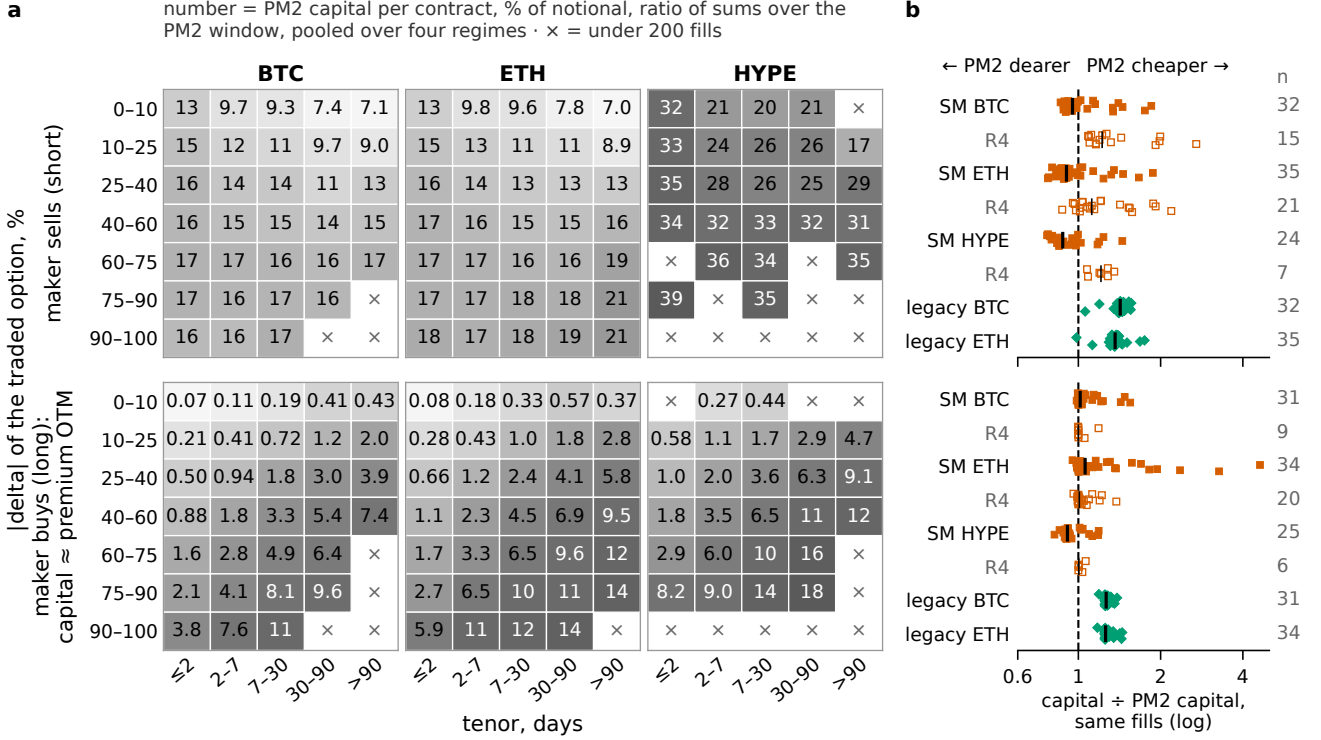


Figure 3: What one contract costs. Panel a is PM2 capital per contract in per cent of notional over the absolute delta of the traded option and tenor, for maker sells (upper row) and maker buys (lower row), as a ratio of sums over the fills of the PM2 window, which the BTC parameter changes of 23 January, 24 May and 20 August 2026 split into four regimes, R1 to R4; the changes of 8 January 2026 and, for HYPE, 8 May 2026 fall inside R1 and R2. Each row is shaded on one logarithmic grey scale, and a cross marks a cell with fewer than 200 fills. Panel b gives, for every occupied cell, the capital of the same fills under standard margin (squares) and under the legacy manager (diamonds) divided by PM2 capital, with the median cell as a bar and the number of cells under n ; the hollow squares of the rows R4 use only the fills after the parameter change of 20 August 2026. For a maker buy, standard margin charges the premium, and PM2 about the premium out of the money (OTM) and less in the money.

same unit per contract. SM covers the whole sample (HYPE from 11 November 2025; six earlier HYPE fills fall outside every registered window), the legacy manager BTC and ETH throughout. The PM2 window is where PM2 covers an underlying: BTC and ETH from 12 June 2025 23:00 UTC, HYPE from 11 November 2025. Only H4 uses events before it.

A fill's market state is rebuilt from the parameter timelines and the last feed values pushed by its block. The fill's stand-alone capital is that of a book holding only the fill, at its price, under every manager available at the block: $q = +1$ for a maker buy and $q = -1$ for a sell (Figure 3). A cell combines an underlying, a maker side and one absolute delta and one tenor bucket of Albiez (2026); a cell with at least 200 fills is occupied. Its edge per unit of capital is a ratio of sums in basis points,

$$e_c^K = 10^4 \frac{\sum_{i \in c} NE_i a_i}{\sum_{i \in c} K_i a_i}, \quad (3)$$

with NE the net edge and a the size; its edge per unit of notional has the index price in place of K .

The maker books are those of the ten subaccounts with the most maker fills, M1 to M10 by rank (not the taker

class of dominant makers in Albiez (2026)). Their opening books are read on chain, with their manager, at the first block of each UTC day. Some share a wallet: M3 with M5, M4 with M10, and M2, the one SM subaccount, with M1, M6 and M8. The book before a fill is the opening book plus the subaccount's balance changes up to the fill's transaction (Addendum 1). The marginal cost of a fill is $\Delta K = K(\text{after}) - K(\text{before})$ per contract under the subaccount's manager, relative to its stand-alone capital. The netting value of a maker-day is the SM capital of its opening book over its PM2 capital, each summed over underlyings, with positions at the companion paper's mark.

For H4 each parameter change j becomes a dose $d_{c,j}$: the mean of $\log(K^+/K^-)$ over the fills of cell c in the fourteen days before the event, where K^+ and K^- are the capital of a fill after and before it in the same market state. Doses are shown in log per cent, 100 times the log change; events whose largest absolute dose is below one log per cent are dropped. Over fourteen days on either side of the event, excluding its day, the half spread (the markout at the fill) in basis points of the index follows

$$y_i = \alpha_{c,j} + \gamma_{t,u} + \beta \text{post}_i d_{c,j} + \varepsilon_i, \quad (4)$$

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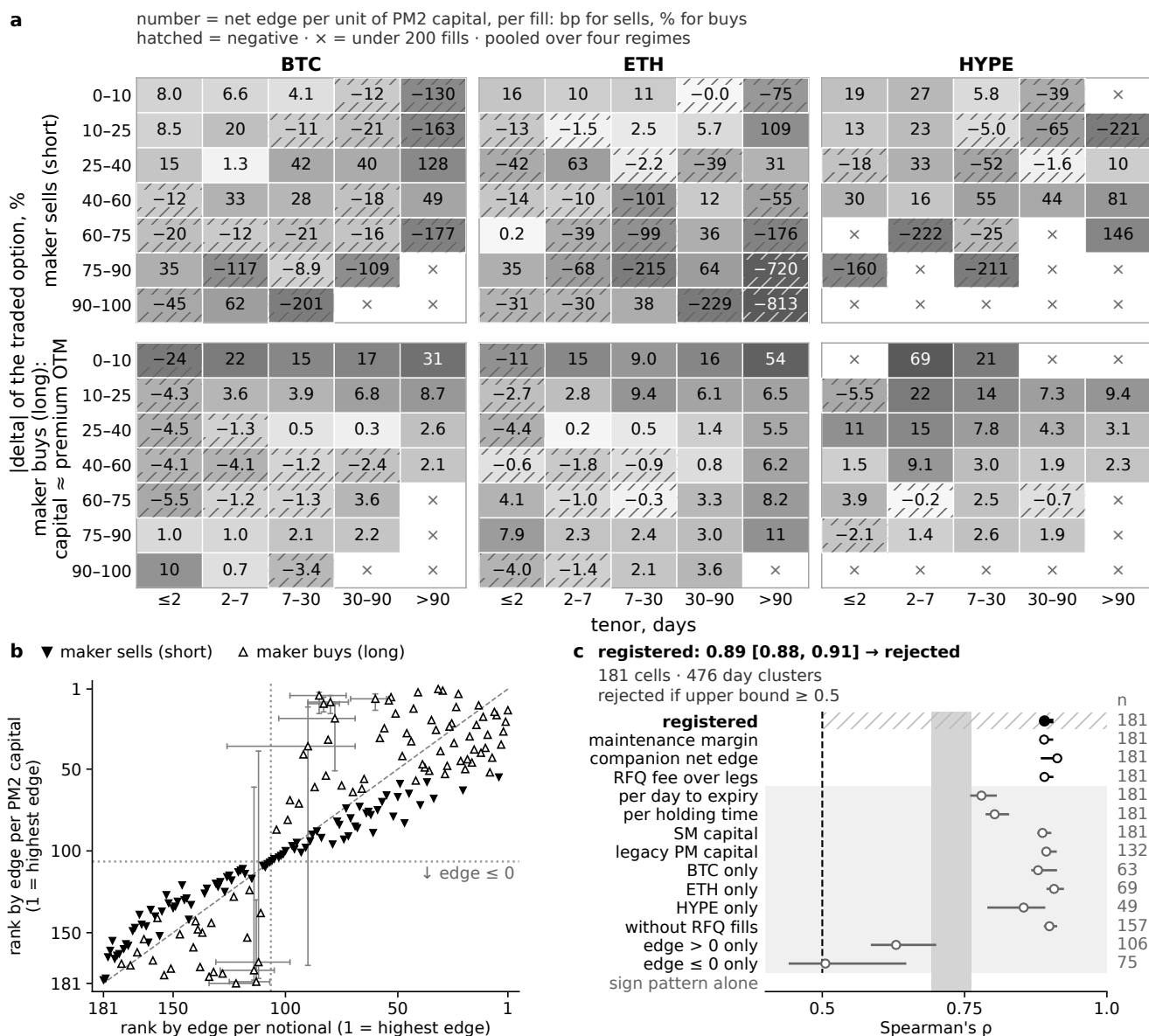


Figure 4: The map in two denominators (H1). Panel a is net edge per unit of PM2 capital, per fill, for every cell of the PM2 window by underlying and maker side, in basis points for maker sells and in per cent for maker buys; negative cells are hatched and a cross marks fewer than 200 fills. For a maker buy the denominator is about the premium when the option is out of the money and less in the money. Darker grey marks a larger absolute value on a log scale, each row on its own. The map per notional is not drawn; its ranks enter panel b. Panel b ranks the 181 occupied cells by edge per notional and by edge per PM2 capital; because capital is positive in every cell, the 106 cells with positive edge come first in both rankings, and the dotted lines mark that boundary. Crosses give the 90 per cent rank intervals of the ten largest moves. Panel c is the registered test: Spearman's ρ with its 90 per cent day-cluster interval (bar, also printed above the panel), set against the threshold of 0.5. The hatched stretch of the registered row, right of the threshold, is the rejection region: the rule rejects H1 if the interval reaches into it. Sensitivities follow and then, on grey, exploratory rows; n is the number of cells in a row, and in the rows by the sign of the edge the cells are chosen again in every replicate. The grey band is the ρ of the sign pattern alone, with ranks shuffled within each sign group.

with post_i marking fills after the event, t the UTC day and u the underlying; cells need 20 fills on each side.

Intervals are 90 per cent percentile intervals of a cluster bootstrap over UTC days with 9999 draws (Cameron et al., 2008). H4 uses a wild cluster bootstrap with Rademacher weights and restricted residuals (MacKinnon and Webb, 2017; Roodman et al., 2019). Its 100 placebo replications draw for each event a date at least 28 days from every

change of the underlying's legacy-manager parameters and default PM2 parameter set (subaccounts may carry their own; Addendum 4). Replicas were validated against the chain's evaluation (eth_call) before any capital entered a test (Appendix A). Analyses not named in the registration are exploratory, those added after the first results post hoc.

registered: 0.035 [0.031, 0.039] → not rejected

4 PM2 subaccounts, ETH and HYPE only
20 000 of 20 000 sampled fills · 385 day clusters

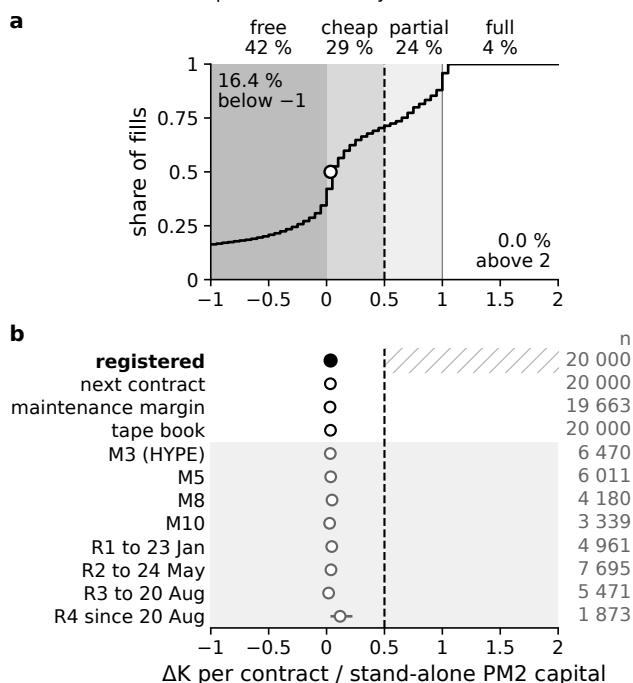


Figure 5: A fill in a dominant maker's book (H2). Panel a is the cumulative distribution of the marginal capital of a fill per contract over its stand-alone PM2 capital, $\Delta K / K_{\text{single}}$, for the 20 000 tested fills of the four PM2 subaccounts, which trade ETH and HYPE only; the numbers above the bands are the shares of fills in each band, and the shares beyond the axis are given at both ends. The dashed line is the registered threshold of one half, and the circle at height one half is the median; its 90 per cent day-cluster interval is narrower than the circle and printed at the top. Panel b repeats the registered row above the sensitivities and, on grey, exploratory rows per subaccount and per regime (R1 to R4). The hatched stretch of the registered row, right of the threshold, is the rejection region: the rule rejects H2 if the interval reaches into it. A maker who starts from an empty book pays the stand-alone capital of Figure 3.

4. Results

4.1. The capital denominator and the map (H1)

Under PM2 a single maker sell binds between 7.1 and 17.3 per cent of notional in BTC, between 7.0 and 21.3 in ETH and between 17.4 and 38.6 in HYPE (Figure 3); a maker buy binds about its premium out of the money and less in the money. Pooled over the window, PM2 is not the cheap manager for one short contract: the ratio of standard margin to PM2 capital in the median sell cell is 0.952 in BTC, 0.905 in ETH and 0.877 in HYPE. After the parameter change of 20 August 2026 the ratio is 1.222 for BTC and 1.120 for ETH, on the 15 and 21 sell cells still occupied. The legacy manager needs more, with medians of 1.423 and 1.363 for BTC and ETH sells.

In the PM2 window 181 cells are occupied, out of 210 with any fill (Figure 4). Across them, Spearman's rank correlation ρ between edge per notional and edge per unit of

registered: 4.73 [4.65, 4.82] → not rejected

9 subaccounts · 1 982 maker-days
475 day clusters (UTC days, not subaccounts)

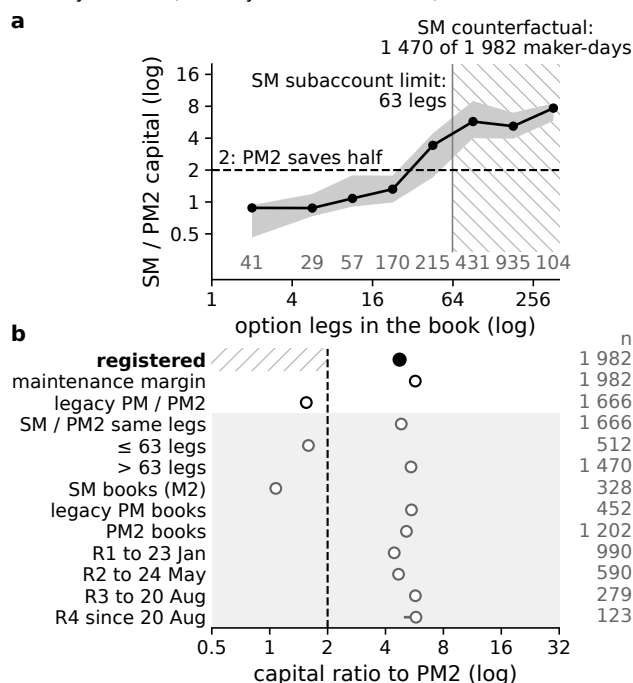


Figure 6: What netting is worth (H3). Panel a is $K_{\text{SM}} / K_{\text{PM2}}$ for the opening books of 1 982 maker-days against the number of option legs, as the median per bin with the interquartile band; the grey numbers above the axis are the maker-days per bin. Beyond 63 legs, the most the endpoint accepted for a standard-margin subaccount in the probes, K_{SM} is counterfactual (hatched); this is the case on 1 470 maker-days. The dashed line is the registered threshold of two. Panel b is the registered median with its 90 per cent interval, the sensitivities including the legacy manager on BTC and ETH legs, and, on grey, exploratory rows: SM against PM2 on the same BTC and ETH legs, by book size, by the manager of the subaccount and by regime. The hatched stretch of the registered row, left of the threshold, is the rejection region: the rule rejects H3 if the interval reaches into it.

PM2 capital is 0.890 (90 per cent interval 0.883 to 0.906); the upper bound exceeds 0.5, so the registered rule rejects H1.

The rejection says less than its margin suggests (post hoc, Appendix B): capital is positive in every cell, so a cell keeps the sign of its edge, and the 106 profitable cells rank first in both maps. Among them alone ρ is 0.630 (0.585 to 0.700), and among profitable buys 0.246 (0.197 to 0.434).

Pooled, the net edge per fill is 3.20 basis points of notional and 38.6 of PM2 capital, not a return per unit of time. The median cell moves by 15 ranks. The largest moves are maker buys far out of the money, whose small capital turns a thin edge per notional into a large edge per unit of capital. BTC buys in the lowest delta bucket with two to seven days to expiry rise from rank 85 to 5; the same buys with at most two days fall from rank 122 to 181, the last. Maker sells, which bind scenario capital, move less, and

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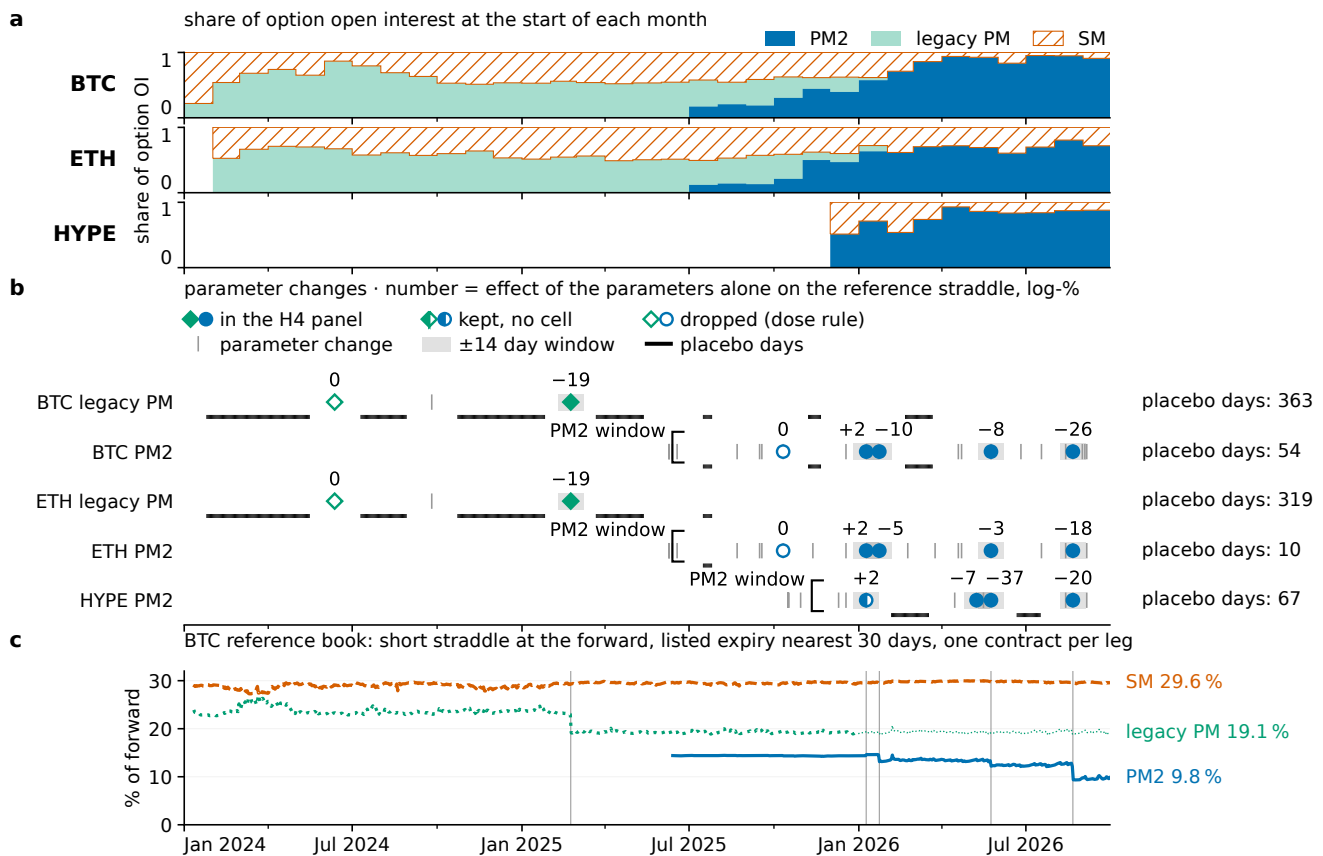


Figure 7: The engine over time. Panel a is the share of option open interest per manager and underlying at the start of each month. Panel b marks every parameter change of the legacy manager and of PM2 as a grey tick and the registered H4 events as symbols: filled when they enter the H4 panel, half filled when kept without a cell of 20 fills on each side, and hollow when dropped by the one log per cent dose rule. The number is the change that the parameters alone make to the capital of the reference straddle of the underlying, in log per cent. Grey bars are the windows of 14 days on either side of each kept event; the windows of January and of May 2026 overlap, so 6527 fills enter two events. Black dashes below each line are the days from which placebo dates may be drawn. Panel c is the capital of the BTC reference book, a short straddle struck at the forward on the listed expiry nearest to 30 days, one contract per leg, in per cent of the forward; the small saw teeth come from rolling between expiries of 21 and 36 days, and the legacy line is thin in months in which the legacy manager held less than 5 per cent of BTC open interest. The grey vertical lines in panel c mark the kept BTC events of panel b. Parameter changes of standard margin left the reference book unchanged.

none ranks higher than 55th per unit of capital. At the top per unit of capital, the edge comes mostly from RFQ legs and is close to the gap between fill price and the stochastic volatility inspired (SVI) mark, which is extrapolated in the far wing (post hoc). The 27 fills with non-positive PM2 capital stay in the sums (Addendum 4).

4.2. A fill in a maker's book (H2)

Of the ten dominant subaccounts, four were margined under PM2 at the start of a day in the window (Addendum 1); they trade ETH and HYPE only. The registered sample draws 20 000 of their 103 961 maker fills. None has non-positive stand-alone capital, the registered exclusion, so all 20 000 enter, on 385 days. Figure 5 shows the distribution of $\Delta K/K_{\text{single}}$, the marginal capital of the whole fill per contract. The median ratio is 0.0346, with a 90 per cent interval from 0.0309 to 0.0386, far below the registered threshold of one half, and 42.1 per cent of the fills leave the

book's capital unchanged or reduce it. H2 is not rejected: in these books the median fill costs 3.5 per cent of what it would cost alone.

The median hides a wide spread and mixes the sides: maker buys mostly release capital and maker sells mostly bind it (post hoc). About a quarter of the fills release more than 0.28 of their stand-alone capital; a quarter bind more than 0.65 of it. From the 95th percentile on, a fill binds about its full stand-alone capital and almost never more. Fills that offset the loss in the book's worst scenario release capital; fills that add to it cost about what they cost alone. Per subaccount (exploratory) the median runs from 0.0264 to 0.0471, and the single HYPE subaccount sits at 0.0323.

4.3. What netting is worth (H3)

Figure 6 reports the capital of the same opening books under SM and PM2. It covers 1 982 maker-days in the PM2

window with at least one option position, from nine subaccounts: four under PM2, four under the legacy manager, for which both capitals are counterfactual, and one under SM; the tenth made no maker fill in the window. No book has non-positive PM2 capital. The median of K_{SM}/K_{PM2} is 4.735, with a 90 per cent interval from 4.646 to 4.819, and eight of the nine subaccount medians exceed two. H3 is not rejected; its registered threshold of two lies at the lower end of the probe-book ratios of Section 2 (2.14 to 10.76 on R). On BTC and ETH legs the legacy manager needs 1.550 times the PM2 capital (a comparison without a threshold), and standard margin 3.234 times the legacy capital.

The factor comes from large books and from the subaccounts that hold them. On 1 470 maker-days (74.2 per cent) the book holds more than 63 option legs, the most an SM subaccount can hold, so its SM capital is counterfactual (Addendum 4). On those days the median ratio is 5.428. On the 512 days with at most 63 legs it is 1.590 (exploratory, interval 1.532 to 1.662), a value the registered rule would reject. But 328 of those days belong to M2, the one SM subaccount. Its books net little (1.076), while its wallet-mates M1 and M8 hold large books. Without M2, the small books give 3.780 (post hoc); books of 32 to 63 legs, M2 included, give 3.43 (Figure 6, panel a). Whether a large book could be split across standard-margin subaccounts is not examined.

4.4. The price of capital (H4)

Figure 7 places the parameter changes next to the capital of a fixed reference book, a short straddle at the forward of each underlying (BTC in panel c). The registered set holds 18 events once same-day changes are merged, without changes that touch only collateral parameters or the number of expiries (Addendum 4); 14 pass the dose filter, and 13 have cells with at least 20 fills on each side, 475 cell-event pairs: the H4 panel. The legacy-manager changes of 22 February 2025 for BTC and ETH, before the PM2 window, supply 75 of them. Of the 14 kept events, 11 made the reference book cheaper, by between 3.4 and 37.4 log per cent across underlyings and between 7.8 and 26.0 for BTC. The largest BTC step, on 20 August 2026, cut the PM2 capital of the straddle by 26.0 log per cent (22.9 per cent). Only the events of 8 January 2026 raised it, by about 1.6 log per cent.

With its fixed effects, β compares, within the same event, cells whose capital changed more with cells whose capital changed less, mostly sells against buys (Figure 8, panel a). It assumes that such cells would have moved in parallel without the change and that makers did not anticipate it; the venue chose the changes itself. The dose is the change in stand-alone capital, while in a dominant book a fill binds far less (H2). The half spread is measured on fills, not quotes. At the kept events, the changed manager held between 54.5 and 94.9 per cent of the open interest.

The estimate of β is -4.60 basis points of the index per unit of log capital, with a one-sided wild cluster bootstrap p of 0.6224, and 64 per cent of the 100 placebo estimates are at least as large (Figure 8). H4 is rejected unless the estimate

is positive with a one-sided p of at most 0.05 and lies above the 95th placebo percentile, here 23.41. The estimate fails both criteria, and H4 is rejected. The sign is the opposite of the hypothesis, and with a cluster standard error of 13.09 the estimate cannot be told apart from zero.

The test has little power. Its standard error holds the event dates fixed, and over the 100 placebo replications the t statistic has a standard deviation of 1.82 instead of one (post hoc). Calibrated on those placebo statistics, the 90 per cent range for β runs from -41.6 to 29.7 . For capital ten per cent cheaper, the change in the half spread ranges from -3.13 to $+4.39$ basis points of the index, against a panel mean of 7.93: a narrowing by up to 39 per cent is not excluded. Only 141 day clusters carry the test, and no variant in Appendix B meets the registered rule.

4.5. Sensitivities

No sensitivity the registration names changes a verdict; maintenance margin is computed for H1 to H3 only. Under maintenance margin, ρ is 0.890 and the netting ratio 5.713. The median marginal cost is then 0.0298 on 19 663 fills with each subaccount's own PM2 parameter set (some carry a lower maintenance factor) and 0.0278 with the default set; using one set for both capitals of a ratio was decided after the first results. With the net edge as defined in the companion paper's first version, ρ is 0.913; with the fee of an RFQ package spread over its legs, it is 0.890. With the book rebuilt from the trade tape, the median marginal cost is 0.0339, and for the next single contract, which the registered wording also covers, 0.0330; Addendum 4 chose the whole fill once both measures existed, before any median was computed.

5. Discussion

For a maker or a quoting bot on this venue the results have practical consequences. First, capital leaves the order of the map largely intact (H1), but from an empty or small book the side matters more than the cell. A long option far out of the money binds about its premium, and a short one binds scenario capital. The same edge per notional is therefore a large edge per unit of capital when the maker buys a cheap wing and a small one when it sells. The top cells per unit of capital owe their edge mostly to RFQ legs and to the SVI mark extrapolated in the far wing (post hoc). Buys mostly release capital in the dominant PM2 books (post hoc), where marginal capital orders the sides differently from stand-alone capital.

Second, the book, not the contract, decides the cost of a quote. In four large PM2 books the median fill binds 3.5 per cent of its stand-alone capital (H2), so stand-alone capital (Figure 3) overstates the median cost of a quote many times over. The endpoint instead returns the marginal capital against the maker's own book in one request.

Third, netting pays for books of a few dozen legs or more held under the portfolio managers (H3; the split by book size is exploratory). The one standard-margin subaccount holds small books that net little; its operator keeps the large books under portfolio margin.

What does the edge cost?

13 events · 475 cell-event pairs · 91 446 rows from 84 919 fills (6 527 in two windows)

141 day clusters · 322 day × underlying effects

panel a: ▼ maker sells (short) · △ maker buys (long) · colour = manager

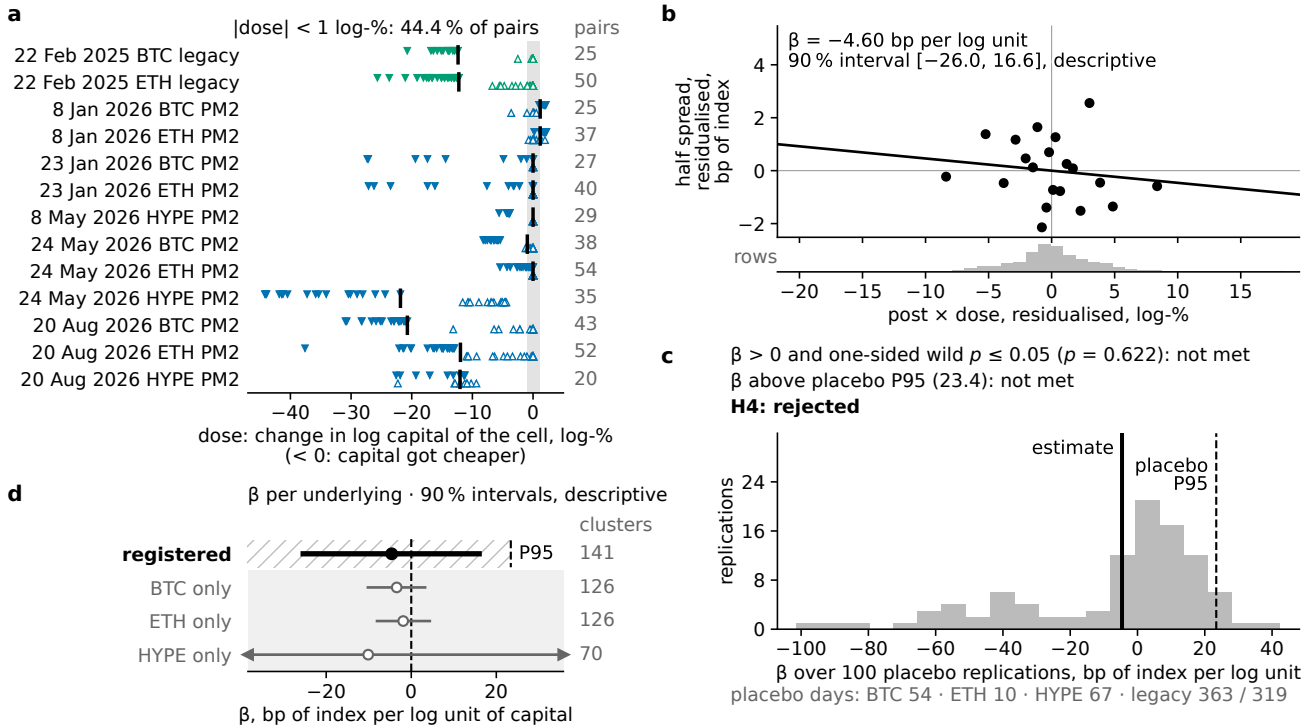


Figure 8: The price of capital (H4). Panel a shows the dose, the change in log capital that a parameter change makes to a cell, for each of the 475 cell-event pairs of the 13 events in the H4 panel, sells above and buys below each line, with the median as a bar. Panel b is the half spread against the post-event dose after removing the cell by event and the day by underlying effects, in 20 bins of equal size, above a grey histogram of that dose over the rows of the H4 panel, one per fill and event; the line shows the estimate β , whose interval is descriptive because the registered p comes from restricted residuals. That day-cluster interval is too narrow, as the placebo replications show; calibrated on them, the 90 per cent range runs from -41.6 to 29.7 (Section 4.4). Panel c places β among the estimates of 100 placebo replications and lists both registered criteria. Panel d gives β per underlying, for exploration; in the registered row, hatching marks the values of β at which the rule rejects H4 whatever its p , up to the 95th placebo percentile (P95) or zero, whichever is larger. Most doses are close to zero or negative, so β is identified from parameter changes that made capital cheaper. Doses on the axes are in log per cent, 100 times the change in log capital, so the line in panel b has the slope $\beta/100$ per log per cent. To read the slope: capital ten per cent cheaper is a dose of -0.105 , or -10.5 log per cent, which predicts a change in the half spread of -0.105β basis points of the index.

Fourth, parameters are a risk the maker does not control: a single change cut the PM2 capital of the BTC reference straddle by 22.9 per cent, and the half spread showed no detectable response (H4). Capital may have been slack, makers may have been bound by something else, or the test lacked power: for capital ten per cent cheaper, a narrowing by up to 39 per cent is not excluded (post hoc, Section 4.4).

Several limits bound these readings. Capital is what the engine would require for the positions, not what makers deposited. Initial margin is assumed to bind. Margin for open orders, charged by the venue but not returned by the endpoint, lies outside the measure, as does every external bound on a book. Capital follows the on-chain engine, while the venue trades on the off-chain one. The net edge deducts a hedging cost, while capital is that of the unhedged contract. H2 rests on four PM2 subaccounts that trade ETH and HYPE only, H3 on books mostly larger than a standard-margin subaccount accepts, and H4 on 13 events that mostly made

capital cheaper. Collateral other than USDC, held on most maker-days, is left outside K .

6. Conclusion

On Derive the capital behind a quote is not a private number. The rules that set it are public smart contracts with parameters on chain, so the capital of any book can be computed at any block, including books nobody held. Measured that way, capital does not reorder the maker's map in the registered sense (H1): the rank correlation is 0.890. Post hoc, much of that correlation comes from the sign of the edge, and the order changes more among profitable cells and at the top of the map. The median fill in a dominant PM2 book binds 3.5 per cent of its stand-alone capital (H2). Netting is worth a factor of 4.73 over standard margin on the books that the dominant subaccounts hold (H3), and most of these books are too large for a standard-margin subaccount. The half spread showed no detectable response

when parameter changes made capital cheaper (H4), but the test has little power. Four hypotheses were registered before any capital figure of the registered analysis existed; H1 and H4 are rejected, H2 and H3 are not. For a maker the capital that matters is that of the next fill in its own book, and the off-chain engine quotes it on request.

Data, code and pre-registration

Every input to this paper is public. The fills come from the venue’s public trade tape as in Albiez (2026); feeds, parameters and maker holdings are read from the smart contracts on Derive Chain by historical `eth_call`; the present-day probes of the off-chain engine use its public endpoint (Derive, 2026a). The code is at github.com/gelatotrader/Derive-Research, together with its tests. It rebuilds the three managers from their smart contracts (Derive, 2026b), validates them against the chain, computes every capital figure, runs the inference and draws every figure. Before publication, an internal audit found errors in the code, the inference, the figures and the text; none of them changes a registered verdict.

The four hypotheses, their rejection rules and the inference were committed in `docs/paper2/PRAEREGISTRIERUNG.md` on 24 September 2026 at 22:33 UTC+2, in commit `c9e9162`, before any capital figure of the registered analysis had been computed (Burlig, 2018). The pre-registration is written in German, and its addenda are headed *Nachtrag*; an English translation made after the analysis is in `docs/paper2/PREREGISTRATION.md`, and the German original is binding. Seven dated addenda follow in the same file; the first four are of 25 September 2026 (UTC+2). Addendum 1 (`b5c905c`) fixes the exact on-chain book for H2 before the first capital figure of the registered form. Addendum 2 (`60896fe`) fixes the unit of fee and rebate, and Addendum 3 (`2416902`) the details of bootstrap and placebo, both before the first inference. Addendum 4 (`a02058a`) records the readings of H1 to H4. Addenda 1 to 3 were each committed before the step they govern. Addendum 4 was written after stand-alone capital, marginal costs, netting ratios, the doses and the H4 panel with its half spreads had been computed; it records readings already implemented, but precedes any test statistic. The validation fixtures committed just before Addendum 1 had already read the on-chain margins of individual maker subaccounts, cash included.

Probes preceded the registration: the probe books of Section 2, which set the threshold of H3; a map of edge per unit of SM capital over the BTC cells, from which the rank correlation that H1 tests could already be read for SM; and margin histories of at-the-money contracts under the three managers, which the registration does not list. The half spreads that enter H4 were known from the companion paper before registration; the registration fixes the doses, the events and the test.

All commit times are local times of the author’s machine. Before publication the history was rewritten twice, the first time to remove account identifiers. The registered texts are

byte-identical to those of the original commits and the commit times are unchanged; Addenda 5 and 7 list the old and the new hashes of every registration commit, and the counts that Addenda 1 to 4 quote refer to the pilot cut. Addendum 5 of 27 September 2026 (`8c066a7`) and Addendum 7 of 5 October 2026 (`7c01e85`), both written after the results on the pilot cut (an earlier cut of the same data, to 17 September 2026), record the rewrites and change no rule, test statistic or result. Addendum 6 of 30 September 2026 (`6f2436e`), written after the results on the pilot cut and before the final data run, keeps the fee of a multi-leg RFQ package on the leg on which the venue books it, as the companion paper does, and adds the fee spread over the package as a sensitivity of H1, under which ρ is 0.890 (Figure 4, panel c). The pre-registration became public with the push of this history on 5 October 2026 (UTC): after the results on the pilot cut, which already held most of the sample and every window of H4, but before the final data run behind every number of the registered analysis. No public timestamp precedes the results on the pilot cut, and the byte identity of the registered texts across the rewrites can be checked only in the author’s local repository. The history shows the order of the commits, not what was computed locally between them.

Competing interest

I am building a market-making bot for the venue studied here and have been in contact with its team about obtaining order-level data for a follow-up paper. No data, funding or review was received from the venue for this paper, and every number in it comes from public sources. Readers should weigh the practitioner conclusions in Section 5 in that light.

Use of generative tools

AI assistants (Claude, Anthropic) supported this work: in searching the literature, in writing and testing code, in cross-checking results and in drafting and editing the text. The research question, the design of the study and the interpretation of the results are the author’s; the author directed the work, reviewed its results and takes full responsibility for the content of this paper.

A. Validation of the replica

Figure 9 compares each offline replica with the deployed smart contracts, on 100 random blocks per underlying and manager (registered: at least 48) and on 20 maker-days. The 877 cases lie at blocks from 13 January 2024 to 30 September 2026, and none reverted on chain. All 32 cells of underlying, manager, kind and margin meet both registered thresholds. For single contracts under initial margin the median absolute relative deviation is 6.8×10^{-10} and the 95th percentile 1.5×10^{-8} ; over all 800 contracts the largest is 8.3×10^{-8} . For 77 evaluations of 26 books from 20 maker-days under the available managers, with 2 to 303 legs, the median is 1.9×10^{-9} and the largest absolute deviation 0.110 USDC. In H3, 10 maker-days hold more legs than

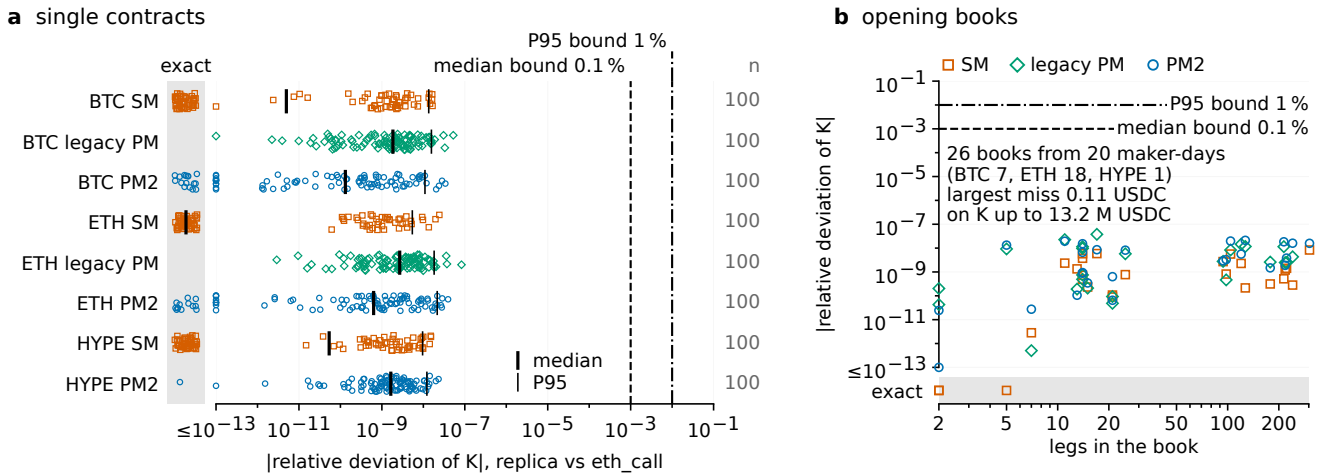


Figure 9: Does the replica match the chain? Absolute relative deviation of initial-margin capital between each offline replica and eth_call on the deployed smart contracts, for single contracts per underlying and manager (panel a) and for the opening books of 20 maker-days against their number of legs (panel b). Exact matches sit in the strip on the left of panel a and at the bottom of panel b; deviations below 10^{-13} sit on the edge of the axis. Thick ticks mark the median of a row, thin ticks its 95th percentile (P95). The lines are the registered bounds for the median (0.1 per cent) and the 95th percentile (1 per cent).

the largest validated book, up to 317; for these the replica is extrapolated in size but not in rule. For HYPE a single maker-day is validated. H2 also prices books larger than any validated one. One validation case, a HYPE contract under standard margin, falls in the settlement window before an expiry, when part of the forward is already fixed, and matches exactly. In the PM2 window no fill uses a volatility feed older than 1 200 seconds or a forward feed older than 3 600 seconds, the feed ages the validation allows; 17 fills use a spot price older than the heartbeat of the spot feed (180 seconds) and stay in the sample.

B. Post hoc and exploratory results

Most of these analyses were added after the first results and are post hoc. The registration lists the maps under other managers, per underlying and per unit of time as exploratory, and the placebo rule over every timeline was computed with the results. None changes a verdict.

For H1, ranks shuffled within each sign group give a ρ of 0.728 on average, with 5th and 95th percentiles of 0.692 and 0.762. Among sells ρ is 0.988 and among buys 0.705. In the groups by the sign of the edge every replicate selects the cells by its own estimated edge; with the cells held fixed, the intervals would ignore that selection. Of the ten best cells per unit of capital, none is among the ten best per notional; of the best 20, five are. The percentile interval of the registered test is not centred on the estimate: 73 per cent of the draws lie at or above it, and a bias-corrected interval runs from 0.871 to 0.898. Maps in the PM2 window under the other managers or per underlying give ρ between 0.854 and 0.907; maps normalised by time to expiry or by the empirical holding period give 0.780 and 0.803 (Figure 4, panel c). Without RFQ fills ρ is 0.899 (0.889 to 0.913) over 157 cells. In the ten best cells per unit of PM2 capital, RFQ fills carry 93.4 per cent of the edge with 39.3 per cent of the fills.

For H2, the median runs from 0.0178 to 0.1172 over the regimes R1 to R4 of Figure 3. By maker side the median is 0.074 (0.062 to 0.094) for sells and -0.128 (-0.202 to -0.062) for buys. Weighted by stand-alone capital, as the sum of ΔK over the sum of stand-alone capital times size, all fills bind 0.137 of their stand-alone capital, sells 0.279, and buys release 0.464. Each leg of a multi-leg RFQ package is measured against the book before the whole package (Addendum 1), without the other legs. Collateral other than USDC, which the registration leaves outside K , is held on most maker-days. One PM2 subaccount holds ETH that its PM2 parameter set counts as offsetting risk, so counting that ETH would lower its PM2 capital.

For H4 the estimate is -3.42 for BTC, -1.90 for ETH and -10.12 for HYPE. It is -3.81 on sell cells alone and -5.13 with the dose weighted by the open-interest share of the changed manager. All of these estimates have a one-sided p above 0.5. On buy cells, whose doses are near zero, it is 128 with a p of 0.13. These p rest on the day-cluster standard error, which the placebo replications show to be too small. A wider reading of Addendum 4 keeps placebo dates at least 28 days from every parameter change of the underlying under any manager, including SM and the subaccounts' own PM2 parameter sets. It gives a 95th placebo percentile of 21.57, and 88 per cent of the placebo estimates are at least as large as β . The placebo panels hold one event more than the real panel, since the HYPE event of 8 January 2026 has no cell there. The placebo dates of ETH under PM2 come from 10 days between 16 July and 25 July 2025, so their windows overlap. With placebos only for events with cells and with disjoint windows per underlying, the 95th percentile is 26.02, 48 per cent of the placebo estimates are at least as large, and the placebo t has a standard deviation of 1.77. The dose is a mean of log ratios, and a few fills near zero capital move it. With the median log ratio as dose the

estimate is -4.52 ; without fills whose log ratio exceeds one in absolute value, it is -4.08 .

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